ABSTRACT

Choosing the right career path has become a challenging task for students and job seekers because of the rapid growth of technology, changing job market requirements, and the large number of available career options. Many students are unsure about which career is suitable for them, what skills are required for a particular job role, and how they should develop those skills in a proper sequence. Traditional career guidance methods mainly provide general suggestions and may not consider an individual's education, interests, existing skills, and career goals together. To address these challenges, this project proposes an **AI-Powered Career Roadmap and Skill Recommendation System** that provides personalized career guidance using Artificial Intelligence and Machine Learning techniques.

The proposed system collects relevant user information such as educational qualification, interests, existing skills, preferred career field, and target job role. Based on this information, the system analyzes the user's profile and recommends suitable career options. It also identifies the skills required for the selected career and compares them with the user's existing skills to determine skill gaps. Based on the identified gaps, the system generates a structured career roadmap that helps the user understand which skills, tools, technologies, courses, certifications, and practical projects should be considered for career development. The system aims to provide recommendations in a simple and understandable manner so that users can make better career decisions.

The proposed system reduces the time and effort required to manually search for career information from multiple sources. By providing personalized recommendations and a structured learning path through a single platform, the system can support students and job seekers in planning their career development. The integration of Artificial Intelligence makes the system more flexible and capable of providing recommendations according to individual user requirements. Overall, the system acts as an intelligent career guidance assistant that helps users understand their career possibilities, identify skill gaps, and follow a systematic roadmap toward their desired career.

INTRODUCTION

Career selection is one of the most important decisions in a student's academic and professional life. With the rapid development of technology and the continuous introduction of new job roles, students have access to a large number of career opportunities. However, having many choices can also create confusion. Students often find it difficult to understand which career path is suitable for their interests, educational background, existing skills, and future goals. They may also lack proper information about the skills and qualifications required to obtain a particular job role.

Traditional career guidance generally depends on teachers, career counsellors, friends, family members, online searches, and general career websites. Although these sources can provide useful information, they may not always provide personalized guidance. A student may know that a particular career requires several technical and soft skills but may not know which skills should be learned first, what their current skill gaps are, or how to create a step-by-step learning plan. Searching for this information manually from different sources can be time-consuming and may result in an unstructured learning process.

Artificial Intelligence provides an opportunity to improve the career guidance process by analyzing user-specific information and generating personalized recommendations. AI and Machine Learning techniques can be used to identify relationships between user profiles, career roles, and required skills. By analyzing information such as educational qualification, interests, existing skills, preferred domain, and career goals, an intelligent system can recommend suitable career options and provide relevant skill suggestions.

The **AI-Powered Career Roadmap and Skill Recommendation System** is developed with the objective of providing personalized career guidance through a single platform. The system takes user information as input and analyzes the user's requirements to recommend suitable career paths. Once a career is selected, the system identifies the important skills associated with that career and compares them with the user's existing skills. This helps in identifying the skills that the user needs to improve or acquire.

One of the major features of the proposed system is the generation of a **personalized career roadmap**. Instead of simply displaying a list of required skills, the system organizes the recommended skills and learning areas into a structured path.

LITERATURE REVIEW

Career guidance and recommendation systems have become increasingly important because students and job seekers often face difficulties in selecting suitable career paths. Traditional career guidance mainly depends on teachers, counsellors, family members, and general online resources. These approaches can provide useful information, but they may not always consider an individual's educational background, interests, existing skills, and career objectives together. The development of Artificial Intelligence and Machine Learning has provided new opportunities for creating intelligent systems that can analyze user information and provide personalized career recommendations.

Several researchers have explored the use of Machine Learning techniques for career prediction and recommendation. These systems generally consider factors such as academic performance, educational qualification, interests, skills, personality characteristics, and user preferences to identify suitable career options. Classification and recommendation algorithms can be used to analyze relationships between user characteristics and different career roles. Such systems can help students understand possible career choices, but many existing approaches mainly focus on recommending a career and provide limited guidance about the steps required to achieve the selected career.

Skill recommendation systems have also been studied to identify the technical and non-technical skills required for specific occupations. These systems can analyze job descriptions, career databases, skill datasets, and user profiles to recommend relevant skills. Skill recommendation is particularly useful because industry requirements are continuously changing and students need to develop skills that are relevant to their desired job roles. However, providing only a list of required skills may not be sufficient for students because they also need to understand which skills they already possess and which skills they need to develop.

Skill-gap analysis is another important area related to career development. It involves comparing the existing skills of an individual with the skills required for a particular job role or career. The difference between the two represents the user's skill gap. Several approaches have used data analysis and Machine Learning to identify these gaps and recommend suitable learning areas. Skill-gap analysis can help users understand their weaknesses and focus their learning efforts on the skills that are most important for their career objectives.

PROBLEM STATEMENT

Selecting a suitable career path is a major challenge faced by students and job seekers. The current job market contains a wide range of career roles, technologies, and skill requirements that change continuously. Because of this, students often find it difficult to determine which career is suitable for their educational background, interests, abilities, and future goals. Many students depend on general internet searches, friends, teachers, or career counselling for guidance, but these methods may not provide personalized and systematic recommendations.

Another major problem is identifying the skills required for a particular career. Even when students know their desired job role, they may not know which technical skills, tools, technologies, soft skills, certifications, and practical experience are necessary to achieve that career goal. They may also be unaware of the difference between their existing skills and the skills expected by the industry. This creates a **skill gap** and makes it difficult for students to prepare effectively for employment.

Existing career guidance platforms generally provide information about careers, courses, or job roles, but many of them do not generate a complete personalized roadmap based on an individual's profile. Users may have to collect information from multiple sources and manually decide what to learn and in what order. This process can be time-consuming, confusing, and inefficient.

Therefore, there is a need for an intelligent system that can analyze a user's educational qualification, interests, existing skills, career preferences, and target career to provide personalized career recommendations. The system should also identify the skills required for the selected career, determine the user's skill gaps, and generate a structured roadmap for skill development.

The **AI-Powered Career Roadmap and Skill Recommendation System** aims to address these problems by using Artificial Intelligence and Machine Learning techniques to provide personalized career guidance and skill recommendations.